

Cooperative Data Analysis in Supply Chains Using Selective Information Disclosure

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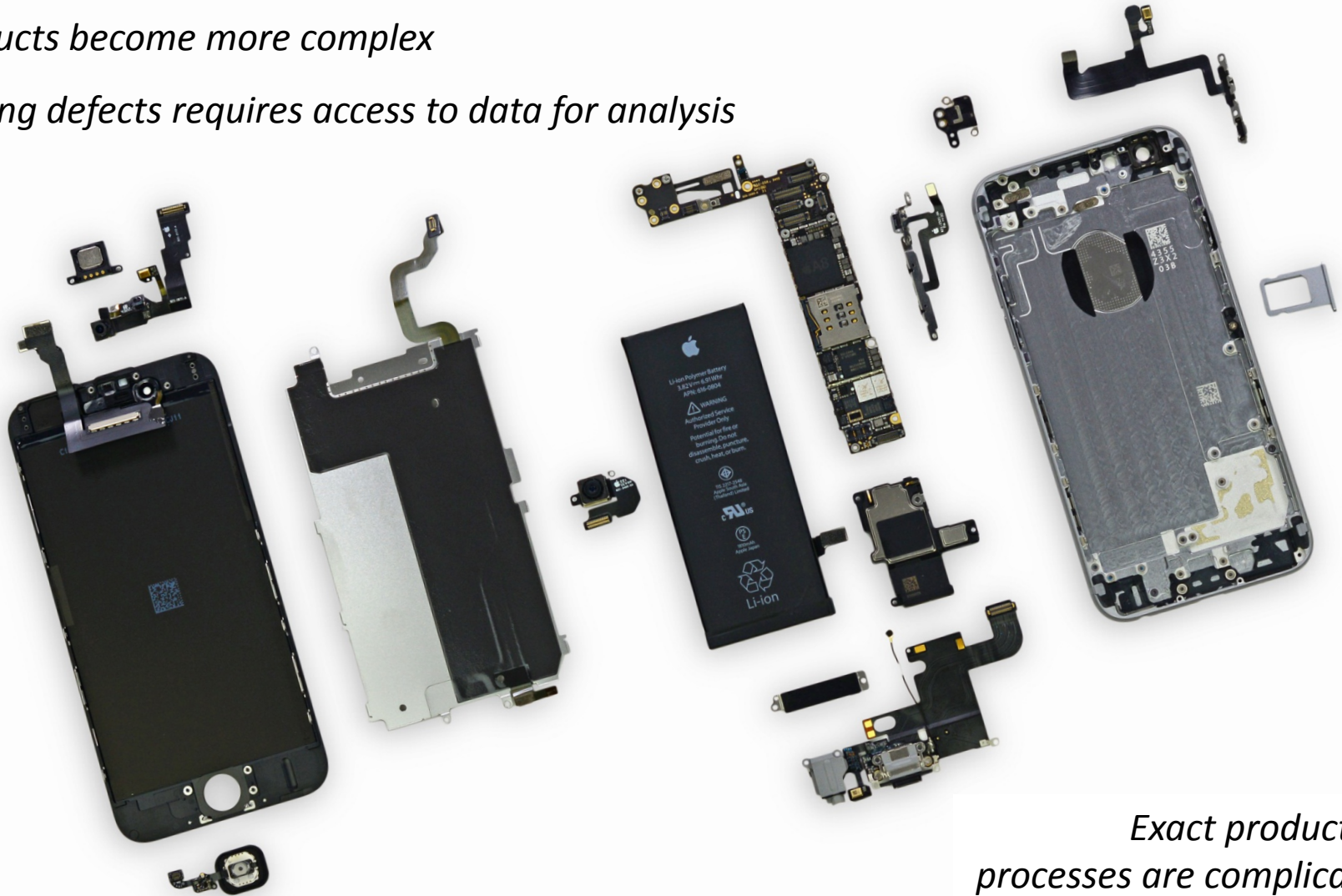


Global supply chain with many suppliers

Products become more complex

Finding defects requires access to data for analysis

Photo: www.ifixit.com



*Exact production
processes are complicated
and may be confidential*

Effective strategies for cooperative data analysis using selective data disclosure.

Background

Privacy preserving data mining

(Aggarwal & Yu 2008, Lindell & Pinkas 2000)

- Protect private information (age, income, etc.)
- Aim: Statistical data analysis on the aggregate

Companies participating in a supply chain

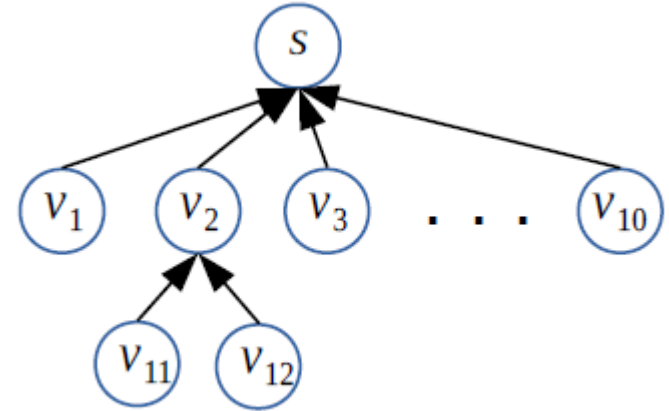
- Incentive to share (mostly logistics) information (Huang, Lau & Mak 2003, Subramani 2004)
- Competition can hinder information sharing (Li 2002, Frohlich 2002)
- Information protection goals are different than for companies
- Root cause analysis (RCA) needs not just logistics information
 - Details about a proprietary production process
 - Change of a third party supplier
 - Large volume of very detailed data

→ Trade-off: Minimize necessary information exchange



Supply Chain

- A directed acyclic graph $G = (V, E)$
- Participants V
- Material and information flows E



Vertically partitioned data set $\mathcal{T}$

Class information
known to s

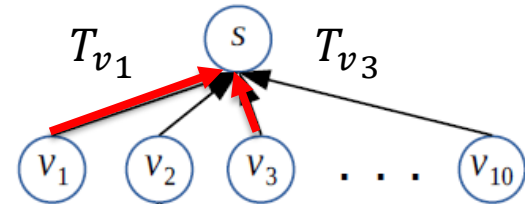
k	T_{v_1}				T_{v_2}			...	$T_{v_{12}}$			c
	$t_{1,1}$	$t_{1,2}$	...	$t_{1,m(v_1)}$	$t_{2,1}$	...	$t_{2,m(v_2)}$	...	$t_{12,1}$	...	$t_{12,m(v_{12})}$	
1	m 1	s 3	...	m 2	s 4	...	e 4	...	e 3	...	m 6	pass
2	m 3	s 9	...	m 2	s 1	...	e 2	...	e 7	...	m 1	fail
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$ K $	m 1	s 4	...	m 1	s 3	...	e 2	...	e 3	...	m 4	pass

Note. m ... machine, s ... supplier, e ... employee.

Protocols for Optimized Information Disclosure

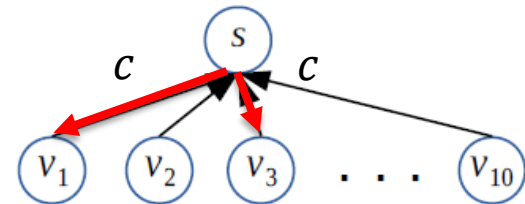
Trivial case

- Single party or full disclosure



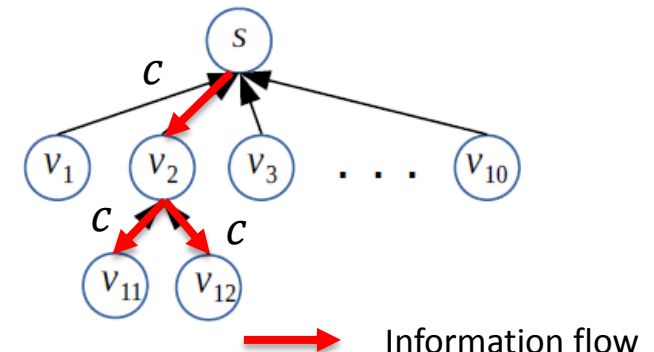
Direct case

- Scenario 1: Known supplier
- Scenario 2: Supplier not known
- Scenario 3: Interaction between suppliers

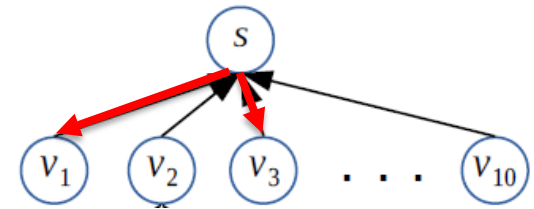


Remote case

- Propagate the class information
- Recursive application of Direct case



Direct Case



Scenario 1: Known supplier

- Supplier v receives class information for analysis.
- Addresses problems and/or reports results to s .

Scenario 2: Supplier not known

- All direct suppliers receive class information for analysis.
- One supplier finds a strong association addresses problem and notifies s .

Scenario 3: Interaction between suppliers

- Like scenario 2, but several supplier find (weaker) associations.
- Further analysis can be coordinated by s .

Analyzing Dependencies

Many methods are available for root cause analysis

Statistical analysis

- Contingency tables and Chi-square test
- (rank) correlation

Data mining (Tan, Steinbach & Kumar, 2006)

- Classification model
 - Decision trees, logistic regression, SVM, etc.
 - Variable importance
- Association analysis
 - Association rule mining

Association Rules

X → class

- Create transactions

k	T_{v_1}				T_{v_2}			...	$T_{v_{12}}$			c
	$t_{1,1}$	$t_{1,2}$	...	$t_{1,m(v_1)}$	$t_{2,1}$	...	$t_{2,m(v_2)}$	...	$t_{12,1}$	...	$t_{12,m(v_{12})}$	
1	m 1	s 3	...	m 2	s 4	...	e 4	...	e 3	...	m 6	pass
2	m 3	s 9	...	m 2	s 1	...	e 2	...	e 7	...	m 1	fail
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
$ K $	m 1	s 4	...	m 1	s 3	...	e 2	...	e 3	...	m 4	pass

Note. m ... machine, s ... supplier, e ... employee.

$$T_{v_1} \bowtie c$$

	$t_{1,1} = m1$	$t_{1,1} = m2$	$t_{1,1} = m3$	$t_{1,2} = s1$	...	$t_{1,m(v_1)} = m1$	c
t_1	1	0	0	0	...	0	1
t_2	0	0	1	0	...	0	0
...	...	...	...	...	...	...	...
$t_{ K }$	1	0	0	0	...	1	1

Association Rules

$X \rightarrow \text{class}$

	$t_{1,1} = m1$	$t_{1,1} = m2$	$t_{1,1} = m3$	$t_{1,2} = s1$	...	$t_{1,m(v_1)} = m1$	c
t_1	1	0	0	0	...	0	1
t_2	0	0	1	0	...	0	0
...	...	...	...	...	...	...	...
$t_{ K }$	1	0	0	0	...	1	1

Support/confidence framework (Agrawal, Imielinski & Swami, 1993)

Set of items: $I = \{ t_{1,1} = m1, t_{1,2} = m2, \dots, t_{1,m(v_1)} = m1 \}$

Left hand side of rule: $X \subseteq I$

Rule: $X \rightarrow c$

Support: $\text{sup}(X \rightarrow c) = \frac{|\{k \in \{1, 2, \dots, |K|\} ; X \cup c \subseteq t_k\}|}{|K|} > \sigma$

Confidence: $\text{conf}(X \rightarrow c) = \frac{\text{sup}(X \cup c)}{\text{sup}(X)} > \gamma$

Association Rules

$X \rightarrow \text{class}$

	X	$\bar{X}$
c	100	3
$\bar{c}$	3000	400000

One-sided Fisher's exact test to measure the strength of rules (Hahsler & Hornik, 2007).
Accept associations with a p-value $< \alpha$

- Correction for multiple comparisons (Miller, 1981)

Bonferroni Correction:

$$\alpha = \frac{\alpha^*}{m}$$

α ... test sig. level
 α^* ... family wise sig.
 m ... # of tests

- Number of shared tuples $n=|I|$ represent a sample.
Upper limit on sample size based on Chernoff bounds
(Zaki et al., 1997):

$$n = \frac{-2\ln(c)}{\tau \epsilon^2}$$

ϵ ... error rate
 $1-c$... confidence level
 τ ... support

Simulation Study

Parameter	Symbol	Value
Number of suppliers	$ V $	10
Features per supplier	$m(v)$	[5, 20]
Values per feature	$\text{range}(t_{v,i})$	[2, 10]
Base defect rate	e_{base}	1%
Feature defect rate	e_{feature}	100% or 5%
Number of shared tuples	$ \Gamma $	1,000 or 10,000
Number of simulation runs	n	100

Scenario 1: Known supplier

- Simple case of Scenario 2

Scenario 2: Supplier not known

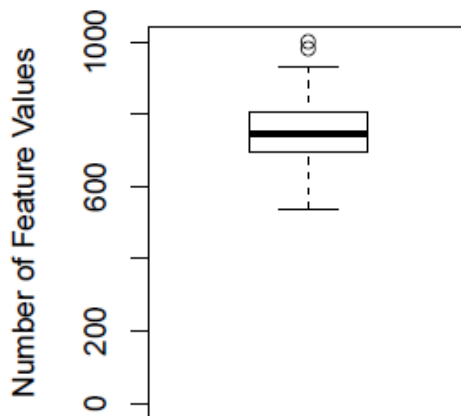
Scenario 3: Interaction between 2 suppliers

- Like Scenario 2, but several supplier find (weak) associations.
- Further analysis can be coordinated by s .

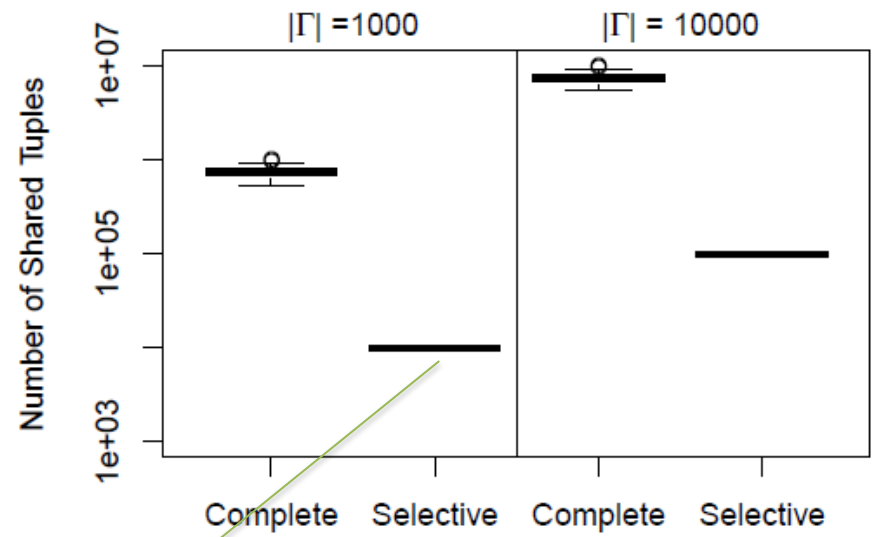
Chernoff bounds
give 240,000 at 1%
support and
confidence and
accuracy level of
95%

Amount of Shared Information

Parameter	Symbol	Value
Number of suppliers	$ V $	10
Features per supplier	$m(v)$	[5, 20]
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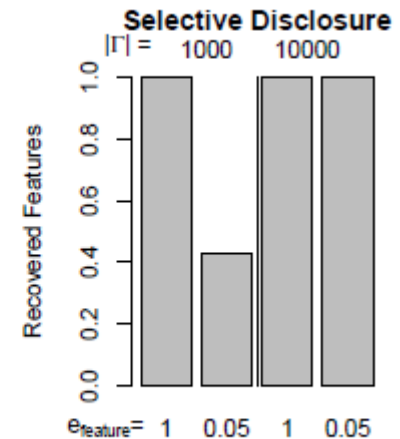
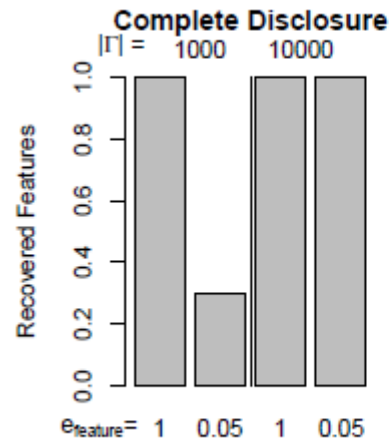
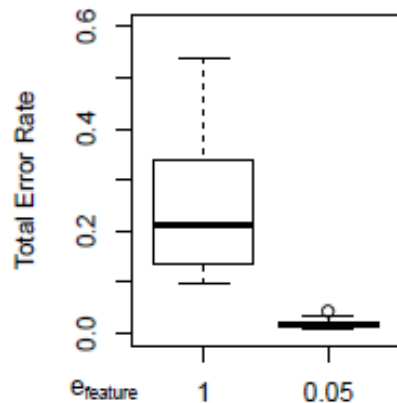
Avg: 744 unique
features values



Disclosure Type

Fixed at $|\Gamma| \times |V|$

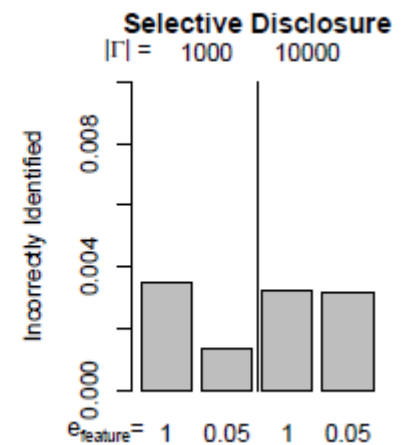
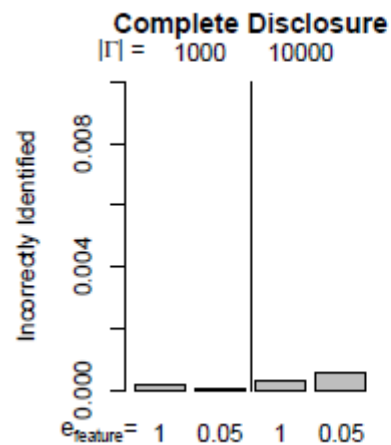
Scenario 2: Supplier not known



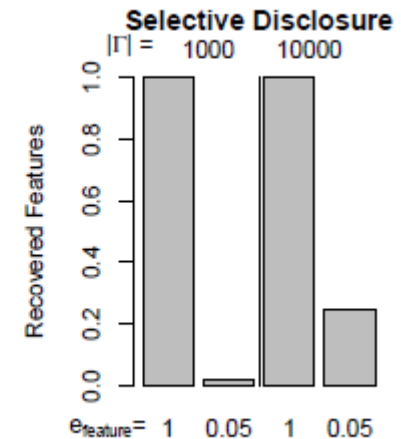
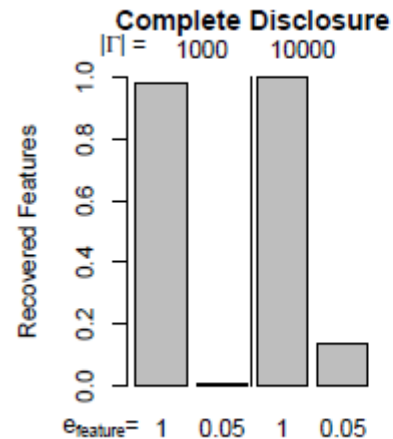
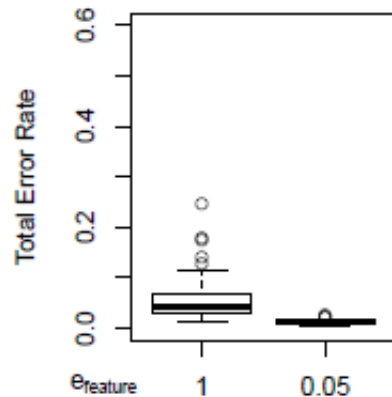
Finding less frequent errors takes more data.

Selective disclosure is as effective as complete disclosure.

Selective disclosure incorrectly reports more features due to undercorrection.



Scenario 3: Interaction between 2 suppliers

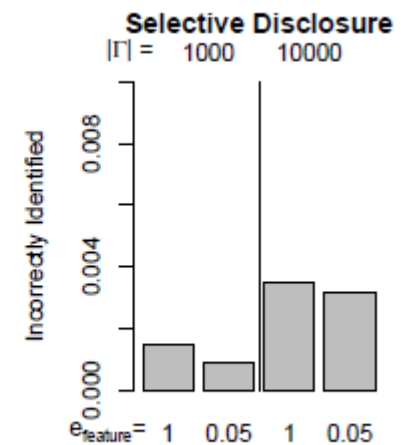
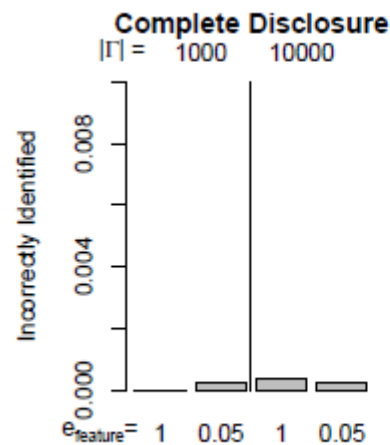


Same as for Scenario 2:

Finding less frequent errors takes more data.

Selective disclosure is as effective as complete disclosure.

Selective disclosure incorrectly reports more features due to undercorrection.



Deployment



<https://rapidminer.com/>

- Easy to use plug-in for RapidMiner.
- Central coordination web service to model supply chain.
- Secure communication directly between participants.
- Participants have full control over what information is shared.



 Direct secure communication

Conclusion

Many modern products are complicated and error-prone.

Data to perform root cause analysis is often not shared in supply chains.

Selective information disclosure:

- Addresses need to performs distributed data analysis
- Minimizes the amount of data to be exposed
- Can be automated such that participants do not need to have in-depth data analysis capabilities
- Initial experiments suggest that it can be effective



Thank you for your attention!

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SMU